

5(a)	<i>any two from:</i> no nucleus ; no organelles / uniform appearance / homogenous cytoplasm ; (some are) biconcave shape / described ; A cells have different shapes same, width / size, as the lumen of the capillary ; A same width as capillary / diameter of cells is 6 - 7 μm R colour / shading / AW	2
5(b)	<i>any three from:</i> 1 (wall is) thin / one cell thick / 1–2 μm in thickness ; R 'thin cell wall' I tunica intima 2 <i>ref. to</i>, endothelial cells / endothelium ; I squamous / epithelium 3 short distance for diffusion ; 4 endothelial pores / fenestrations or pores / gaps, between / within, (endothelial) cells of the wall ; 5 for passage of, (named) small molecules / AW, from / to, plasma / blood / tissue fluid ; A phagocytes A ultrafiltration 6 AVP ; <i>ref. to</i> pinocytosis across endothelial cells	3

Question	Answer	Marks
5(c)(i)	<i>mark to max 3 if no ref. to 'more' at least once in the response</i> <i>any four from:</i> 1 more carbon dioxide diffuses into red blood cells ; 2 more carbonic acid is formed by carbonic anhydrase ; 3 formation of more hydrogen ions ; A H^+ 4 haemoglobin has a high affinity for hydrogen ions ; A H^+ 5 haemoglobin binds more hydrogen ions to form, haemoglobinic acid (HHb) ; A H^+ 6 (formation of HHb) decreases affinity of haemoglobin for oxygen ; 7 haemoglobin releases more oxygen ; 8 carbon dioxide binds to $-NH_2$ / N terminal, of, globin / polypeptides / α chains and β chains ; 9 forms carbaminohaemoglobin ; R carboxyhaemoglobin <i>also lowers affinity</i> 10 allosteric effect / change in tertiary structure / AW, in (oxy)haemoglobin (causes, release / AW, more oxygen) ;	4

Question	Answer	Marks
5(c)(ii)	any two from: 1 hydrogencarbonate ions / HCO_3^- , pass out of red blood cells (into the plasma to increase concentration in deoxygenated blood) ; 2 chloride ions pass into red blood cells ; 3 to replace the, negatively-charged ions / anions / HCO_3^- ; A to achieve electroneutrality / balance the electrical charge 4 chloride shift ; 5 AVP ; chloride ions pass through, channel proteins / anion exchangers / transport proteins / by facilitated diffusion	2

Question	Answer	Marks
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